CBE 141 : Cheat Sheet

known variables = total # variables − # equations

+ Phase Rule

- Gibbs: $F = 2 - \pi + C$
- Duhem: $F = 2 = [2-(N-1)\pi] - [(\pi-1)(N)]$

+ Residual

$$M^R(T,P) \equiv M(T,P) - M^{IG}(T,P)$$

$$\frac{G^R(T,P)}{RT} = \int_0^P \frac{Z(T,P)}{P}\,dP \qquad \frac{S^R}{R} = \frac{(S^R)^0}{R} + \frac{\omega (S^R)'}{R}$$

$$\frac{S^R}{R} = -\int_0^P \frac{(Z-1)}{P}\,dP - T\int_0^P \left(\frac{\partial Z}{\partial T}\right)_P \frac{dP}{P}$$

$$\frac{H^R}{RT} = -T\int_0^P \left(\frac{\partial Z}{\partial T}\right)_P \frac{dP}{P}$$

$$G_i^R = G_i - G_i^{IG} = RT \ln \frac{f_i}{P} \qquad \ln \phi_i = \int_0^P \frac{(Z-1)}{P}\,dP \qquad \Delta H^{IG} = \int_{T_1}^{T_2} C_p(T)\,dT$$

$$\Delta H = \Delta H^R + \Delta H^{IG} \qquad \Delta H^R = H_2 - H_1$$

$$S_2 - S_1 = S_2^R + (S_2^{IG} - S_1^{IG}) - S_1^R$$

+ Equilibrium

$$G^\alpha = G^\beta \qquad \frac{dP^{sat}}{dT} = \frac{\Delta H^{\alpha\beta}}{T\Delta V^{\alpha\beta}}$$

$$\ln\left(\frac{P_2^{sat}}{P_1^{sat}}\right) = -\frac{\Delta H^{vap}}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

$$\mu^\alpha = \mu^\beta$$

+ VLE (Pure Species)

$$G_i^l = \Gamma_i(T) + RT \ln f_i^l$$

$$G_i^v = \Gamma_i(T) + RT \ln f_i^v$$

$$G_i^v - G_i^l = RT \ln \frac{f_i^v}{f_i^l}$$

$$\phi_i^v = \phi_i^l = \phi_i^{sat}$$

$$f_i = y_i P \qquad \phi^{sat} = \frac{f^{sat}}{P^{sat}} = \frac{\hat{f}_i}{y_i P}$$

$$\ln \phi_i = \int_0^P \frac{(Z-1)}{P}\,dP$$

$\phi < 1$ Attractive forces dominate

+ Species in a solution

$$\hat{f}_i^\alpha = \hat{f}^\beta \ldots = \hat{f}_i^\pi$$

$$\bar{G}^R = RT \ln \phi_i$$

$$M = (1-x)M^l + xM^v$$

Quality = x = mole fraction of vapor

$$\hat{\phi} = \frac{\hat{f}_i}{y_i P}$$

+ Partial Molar Prop.

$$d(nG) = (nV)\,dP - (nS)\,dT + \sum_i \mu_i\,dn_i \qquad \mu_i^\alpha = \mu_i^\beta$$

$$M = \sum x_i \bar{M}_i \qquad \sum x_i\,d\bar{M}_i = 0$$

$$\bar{M}_1 = M + x_2 \frac{dM}{dx_1} \qquad \bar{M}_2 = M - x_1 \frac{dM}{dx_1} \qquad \frac{dM}{dx_1} = \bar{M}_1 - \bar{M}_2$$

+ Ideal gas

$$\rightarrow \frac{RT}{P^T} = x_i\left(\frac{RT}{P_i^\pi}\right) + x_j\left(\frac{RT}{P_j^\pi}\right)$$

$$\rightarrow G_i^{IG} = \Gamma_i(T) + RT \ln P$$

$$\rightarrow \mu_i^{IG} = \bar{G}_i^{IG} = RT\ln(y_i P) + \Gamma(T)$$

$$\rightarrow g - g_i^\circ = \mu_i - \mu_i^\circ = RT \ln\left[\frac{P}{P_\theta}\right] = \int_{P_{low}}^P \bar{V}\,dP$$

$$[V = x_1\bar{V}_1 + x_2\bar{V}_2]$$

$$\bar{H}^{IG} = H_i^{IG} = \sum y_i H_i^{IG}$$

$$S^{IG} = \sum y_i S_i^{IG} - R\sum y_i \ln y_i$$

$$G^{IG} = \sum y_i G_i^{IG} + RT \sum y_i \ln y_i$$

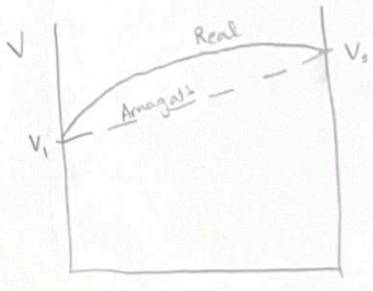
+ Joule Thompson

$\Delta H = 0$, $W_s = 0$

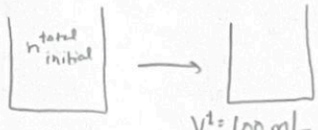
$$\left(\frac{\partial T}{\partial P}\right)_H = \mu_{JT} = \frac{1}{C_p}\left(V - T\left(\frac{\partial V}{\partial T}\right)_P\right)$$

Amagat's law:

+ Straight line connecting V_1 & V_2
+ $V = \sum_i^C x_i V_{i,pure}$



$V = f(x_i)$



7% mol — 15% mol

$$n^t = \frac{V^t}{V(x_i)}$$

n_1 n_2 (in final)

Whole point is to find total moles in initial solution.

$n^{total}_{initial} \cdot V(x_i)$ = Volume

$\frac{mol}{mol} \cdot \frac{mL}{mol} = mL$

$$S_2 - S_1 = \cancel{S_2^R + (S_2^{IG} - S_1^{IG}) - S_1^R}$$

0

$$\rightarrow S^E = \Delta S^{mix} + \underbrace{R\sum x_i \ln x_i}_{\Delta S^{id}} \quad , \quad \Delta S^{id} = -$$

G	P	H
T		S
A	V	U

$dU = TdS - PdV$

$dH = TdS + VdP$

$dA = -PdV - SdT$

$dG = VdP - SdT$

$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V$ $\quad \left(\frac{\partial P}{\partial T}\right)_V = \left(\frac{\partial S}{\partial V}\right)_T$

$\left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P$ $\quad \left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P$

$\Delta U = Q + W$

$W = -\int P dV$

$W > 0$ in compression (done on the system)

$\gamma = \frac{c_p}{c_v}$ $\quad PV^\gamma = P^{1-\gamma}T^\gamma = TV^{\gamma-1} = \text{constant}$

$\Delta H = \int_{T_1}^{T_2} c_p dT$ $\quad \Delta U = \int_{T_1}^{T_2} c_v dT$

$dS^t = \frac{\delta Q_{rev}}{T}$ $\quad \Delta S_{univ} = \Delta S^t_{sys} + \Delta S_{surr}$ ($\Delta S_{univ} = 0$ for reversible process)

$\rightarrow c_p = \left(\frac{\partial S}{\partial T}\right)_P \cdot T = \left(\frac{\partial H}{\partial T}\right)_P$

$\left(\frac{\partial P}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P \left(\frac{\partial T}{\partial P}\right)_V = -1$ $\quad \left(\frac{\partial V}{\partial P}\right)_T = \frac{1}{\left(\frac{\partial P}{\partial V}\right)_T}$

$\rightarrow c_v = \left(\frac{\partial S}{\partial T}\right)_V \cdot T = \left(\frac{\partial U}{\partial T}\right)_V$ $\quad \rightarrow \Delta S = c_p \ln\left(\frac{T_2}{T_1}\right)$ $\quad c_p = c_v + R$ (Ideal gas)

$\left(\frac{\partial P}{\partial V}\right)_T = 0$

Air: 79% N_2, 21% O_2

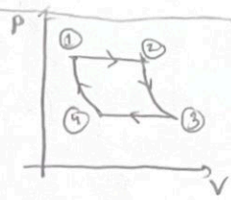
† Energy balance:

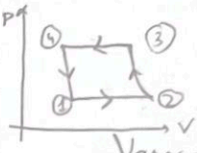
$\Delta H + \frac{\Delta u^2}{2} + g\Delta z = Q + W_s$ (Open S.S.)

$\frac{d(mU)_{cv}}{dt} + \Delta\left[\left(H + \frac{1}{2}U^2 + zg\right)\dot{m}\right] = \dot{Q} + \dot{W}$ (Transient)

† Cycles

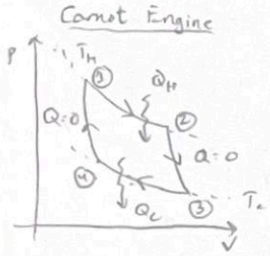
Heat pump



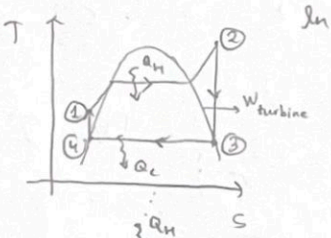


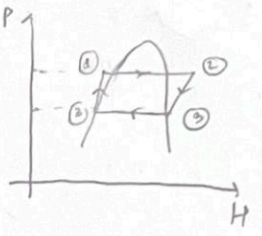
Vapor-Compression

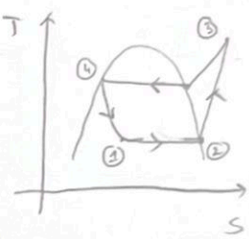
Carnot Engine

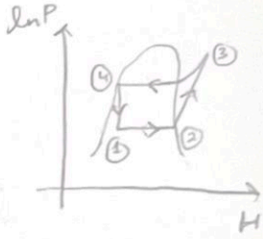


Rankine







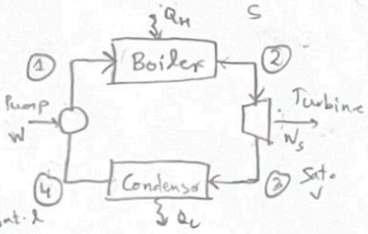


$W = Q_H - Q_L$

$\eta = \frac{W_{net}}{Q_H} = 1 - \frac{Q_L}{Q_H}$

Q=0, $\Delta U = -W$, $\ln\frac{V_a}{V_b} = -\ln\frac{V_d}{V_c}$ $\quad \eta = 1 - \frac{T_L}{T_H}$

$\Delta U = 0$, $\frac{q_H}{q_L} = \frac{T_H \ln(V_a/V_b)}{T_L \ln(V_c/V_d)}$, $\frac{Q_H}{Q_L} = \frac{T_H}{T_L}$

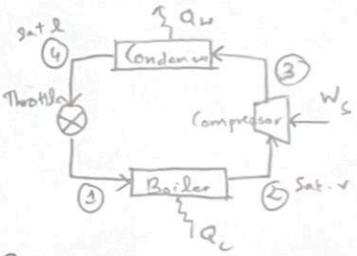


$\eta = \frac{\text{Desired Output}}{\text{Price you have to pay}} = \frac{W_{net}}{Q_H} = \frac{h_3 - h_2}{h_2 - h_1}$

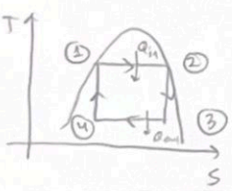
$\Delta S_{surr} = -\frac{q_H}{T_H} - \frac{q_L}{T_L} = 0$

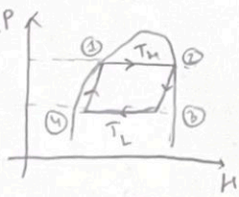
Tons of refrigeration (Power) rating $= \dot{Q}_L$

$\dot{m} = \frac{\text{Power}}{Q_L} = \frac{\dot{Q}_L}{q_L}$ $\quad COP = \frac{Q_L}{W_{net}} = \frac{h_2 - h_1}{h_3 - h_2}$



1 F = 460 Rankine



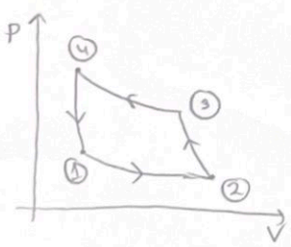


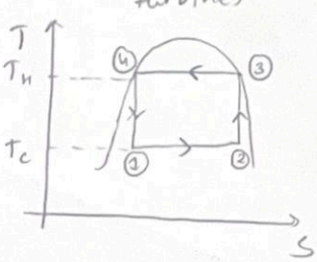
$\eta^{thermal} = \eta^{thermal}_{ideal\ cycle} \cdot \eta^{mech}_{turbine}$

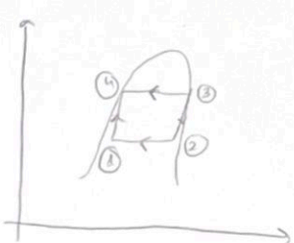
Turbine (expansion): $\eta = \frac{W_{actual}}{W_{ideal}}$

Pump (compression): $\eta = \frac{W_{ideal}}{W_{actual}}$

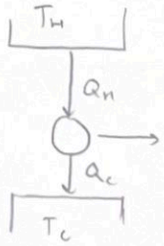
Carnot Refrigerator (Replacing throttle w/ turbine)





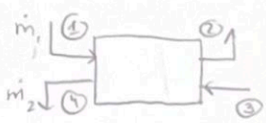


$COP = \frac{T_L}{T_H - T_L}$





Energy balance
Heat Ex



$\dot{m}_1(h_2 - h_1) = \dot{m}_2(h_4 - h_3)$

$\rightarrow S^t = \Delta S^{mix} + \underbrace{R\Sigma x_i \ln x_i}_{\Delta S^{id}} = -$

+EOS

$T_r = T/T_c$ $P_r = P/P_c$

+ Van der Waal: $P = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$ $a = \frac{27}{64}\frac{(RT_c)^2}{P_c}$ $b = RT_c/8P_c$

+ Redlich Kwong: $P = \frac{RT}{V_m - b} - \frac{a}{V_m(V_m+b)}$

$a = \frac{\Psi \alpha R^2 T_c^2}{P_c}$ $b = \Omega \frac{RT_c}{P_c}$ Ψ & Ω are Eos $\alpha = T_r^{-1/2}$

+ 2nd virial: $Z = Z^0 + \omega Z'$

$B^0 = 0.083 - \frac{0.422}{T_r^{1.6}}$ $Z^0 = 1 + B^0\frac{P_r}{T_r}$; $Z' = B'\frac{P_r}{T_r}$

$B' = 0.139 - \frac{0.172}{T_r^{4.2}}$ $Z = 1 + \frac{BP}{RT} = 1 + BP_r$ $T_r = \frac{T}{T_c}$ $P_r = \frac{P}{P_c}$

Random

Moller (H-S) → Isobars $H(S,P)$

① $dH = TdS + VdP \Rightarrow dH = TdS \Rightarrow \left(\frac{\partial H}{\partial S}\right)_P = T$

$\left(\frac{\partial^2 H}{\partial S^2}\right)_P = \left(\frac{\partial T}{\partial S}\right)_P$; $\left.\frac{\partial T}{\partial S}\right|_P = \frac{1}{C_P} \Rightarrow \left(\frac{\partial^2 H}{\partial S^2}\right)_P = \frac{T}{C_P}$

② $U(T,V) \Rightarrow dU = \left(\frac{\partial U}{\partial V}\right)_T dV + \left(\frac{\partial U}{\partial T}\right)_V dT$

$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial S}{\partial V}\right)_T - P\left(\frac{\partial V}{\partial V}\right)^1$ (@ const T)

$dU = TdS - PdV \rightarrow dV$

③ $S(T,P)$ $dS = \left(\frac{\partial S}{\partial T}\right)_P dT + \left(\frac{\partial S}{\partial P}\right)_T dP$

$\left(\frac{\partial S}{\partial T}\right)_P = \frac{C_P}{T}$ $\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P = -\frac{R}{P}$

$\therefore \ln\left(\frac{T_2}{T_1}\right) = \frac{R}{C_P}\ln\left(\frac{P_2}{P_1}\right)$

④ $S(T,V)$

+Excess

$M^E = \Delta M - \Delta M^{id}$ $H^E = \Delta H^{mix}$ $\Delta H^{id} = \Delta V^{id} = 0$ $V^E = \Delta V^{mix}$

$\rightarrow \frac{-H^E}{RT^2} = \frac{\partial}{\partial T}\left(\frac{G^E}{RT}\right)\Big|_{P,x}$

$\rightarrow G^E = \sum x_i \bar{G}_i^E = RT\sum x_i \ln\gamma_i$

$\Delta G^{id} = RT\sum x_i \ln x_i$

$\rightarrow \bar{G}_i^E = \bar{G}_i - \bar{G}_i^{id} = RT\ln\gamma_i$

$\rightarrow \Delta G_{mix} = g - \sum x_i g$; $\Delta G_{mix}^{id} = RT\sum x_i \ln x_i$

$g = \sum x_i g_i + \underbrace{RT\sum x_i \ln x_i}_{\text{ideal}} + g^E$

$\rightarrow S^E = \Delta S^{mix} + \underbrace{R\sum x_i \ln x}_{\Delta S^{id} = -}$

- $\bar{H}_1^E = H^E + x_2 dH^E/dx_1$
- $\bar{H}_2^E = H^E - x_1 dH^E/dx_1$

$\frac{\gamma_a}{\gamma_b} = \frac{P_b^{sat}}{P_a^{sat}}$

+VLE

+ Lewis Randall: $\hat{f}^L = x_i f_{i,Pure}$; $\hat{f}^V = x_i f_{i,Pure}$

$\gamma_i = \frac{\hat{f}^l}{\hat{f}^{Id}} = \frac{\hat{f}^l}{x_i f_i^0}$; Henry's Law: $y_i P = x_i H$ $x \approx 1$

Raoult's Law: $y_i P = x_i P_i^{sat}$

→ Comparing same species in a VLE Mixture: $\hat{f}^V = \hat{f}^L$

$\hat{\phi} y_i P = \gamma_i x_i \phi_i^{sat} P_i^{sat} \exp\left[\frac{V_i^L(P - P_i^{sat})}{RT}\right]$

① Ideal gas + Ideal solution (@ low P)

$\phi^{sat} = \gamma = P = 1$ $y_i P = x_i P_i^{sat}$ (Raoult's Law)

② Ideal gas + Non-Ideal solution (@ low P)

$\phi^{sat} = 1$, $P = 1$ $y_i P = \gamma_i x_i P_i^{sat}$ (Modified Raoult's Law)

③ Non ideal gas + Non-ideal liquid (moderate P)

$P = 1$ $\hat{\phi}_i y_i P = \gamma_i x_i \phi_i^{sat} P_i^{sat}$

$\hat{\phi}^V = \frac{\hat{f}^V}{y_i P}$; $\phi^L = \frac{\hat{f}^L}{x_i P}$ $\ln\phi^{sat} = \frac{B_{11}P^{sat}}{RT}$

$\ln\phi_i^V = \frac{P(B_{11} + y_2^2 S_R)}{RT}$

$S_{12} = 2B_{12} - B_{11} - B_{22}$

→ Consistency:

Gibbs/Duhem → $\sum x_i \ln\gamma_i = 0$

$x_1\left(\frac{\partial \ln\gamma_1}{\partial x_1}\right) + x_2\left(\frac{\partial \ln\gamma_2}{\partial x_1}\right) = 0$

+Mixing

① Van der Waal Mixing Rule: $a_{mix} = \sum_i y_i a_i^{1/2}$ $b_{mix} = \sum_i y_i b_i$ } Use in Van der Waal EOS

② 2nd Virial Pitzer: $B_{ij} = \frac{RT_{c,ij}}{P_{c,ij}}(B^0 + \omega_{ij}B')$; $\omega_{ij} = \frac{\omega_i + \omega_j}{2}$; $P_{cij} = \frac{Z_{cij}RT_{cij}}{V_{cij}}$

$T_{cij} = (T_{ci}T_{cj})^{1/2}(1 - k_{ij})$; $Z_{cij} = \frac{Z_{ci} + Z_{cj}}{2}$; $V_{cij} = \left(\frac{\sqrt[3]{V_{ci}} + \sqrt[3]{V_{cj}}}{2}\right)^3$

$B_{overall} = y_1^2 B_{11} + 2y_1y_2B_{12} + y_2^2 B_{22}$

③ 1 parameter Margules: $\frac{G^E}{RTx_1x_2} = A$

④ 2 parameter Margules: $\frac{G^E}{RTx_1x_2} = A_{21}x_1 + A_{12}x_{12} = x_1\ln(\gamma_1) + x_2(\ln\gamma_2)$

$\lim_{x_2 \to 1} \frac{G^E}{x_1x_2RT} = \ln(\gamma_2)^{x_2=0} = A_{21}$

$\ln(\gamma_1) = x_2^2\left[A_{12}T\,2(A_{21} - A_{12})x_1\right]$

$\ln(\gamma_2) = x_1^2\left[A_{21} + 2(A_{12} - A_{21})x_2\right]$

⑤ Van Laar: $\ln\gamma_1 = A_{12}\left(\frac{A_{21}x_2}{A_{12}x_1 + A_{21}x_2}\right)^2$ $\ln(\gamma_2) = \left(\frac{A_{12}x_1}{A_{12}x_1 + A_{21}x_2}\right)^2 \cdot A_{21}$

+Reaction

$K(T) = \exp\left(\frac{-\Delta G_i^0(T)}{RT}\right)$; $K(T) = K_y K_\phi \left(\frac{P}{1\,bar}\right)^{\Delta\nu}$; $K_y = \frac{y_C^c}{y_A^a y_B^b}$

$K(T) = \prod\left(\hat{f}_i/f_i^0\right)^{\nu_i}$

$\frac{dn_i}{\nu_i} = d\xi$ $\frac{dG}{d\xi} = \sum \mu_i \nu_i = 0$ (@ VLE) $n_f = n_i + \nu_i\xi$

$\frac{d\ln(K)}{dT} = \frac{\Delta H}{RT^2} \Rightarrow \ln\left(\frac{K_2}{K_1}\right) = \frac{-\Delta H_{rxn}}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$